Imade **Bouftini**

PhD Student @ CEA Grenoble · Physics-informed ML for Battery Ageing

Grenoble, France

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PhD student at CEA Grenoble on physics-informed machine learning for battery ageing. Engineer from École Centrale de Nantes (data science, signal processing, applied mathematics), with two deep learning internships in medical imaging: stroke segmentation from 4D CT perfusion at GE HealthCare and breast cancer detection at Ai Movement, from model design to C++ deployment.

Experience

CEA Grenoble, with Université Grenoble Alpes and Sorbonne Université — *Grenoble, France*

PhD Student, Machine Learning for Battery Ageing — 2026 – Present

- Develop physics-informed models of Li-ion and Na-ion battery ageing to estimate state of health and remaining useful life.

GE HealthCare — *Buc, France*

Deep Learning Research Intern (Master's Thesis Internship) — Apr. 2026 – Sep. 2026

- Built an end-to-end deep learning pipeline that segments the ischemic core and penumbra in stroke from 4D CT perfusion, replacing a classical deconvolution method that fails on edge cases and needs manual clinician input.
- Compared with 3 clinical software tools (RAPID, icobrain cva, CTP4D) on 3 public datasets; ablated each architectural choice.
- Mean Dice of **0.63** versus 0.40 for the learned 4DMJNET benchmark; inference in **15 s per case** on GPU instead of 1–2 min.
- Ported the full inference pipeline to C++ with OpenVINO and built a Flask web app for clinical visualization of the results.

Ai Movement — *Rabat, Morocco*

Data Science Research Intern — Apr. 2025 – Aug. 2025

- Implemented the multi-view *Act Like a Radiologist* (AGN) architecture for breast cancer detection in PyTorch from the paper alone, with no reference code, on scarce clinical data.
- Redesigned the attention gates (weights in $[-1, 1]$), added residual connections, trained in two stages (Mask R-CNN, AGN).
- Improved Recall@0.5FPI by **9.5 points** over the Mask R-CNN, DETR and YOLOv8 baselines.

Education

École Centrale de Nantes — *Nantes, France*

Diplôme d'Ingénieur (Master's Degree in Engineering) — Sep. 2023 – Sep. 2026

- **Specialization II, Mathematics & Applications** (2025–26, GPA 3.85/4.0)
- **Specialization I, Data Science & Signal Processing** (2024–25, GPA 3.99/4.0)

Lycée Mohammed VI d'Excellence — *Benguerir, Morocco*

Preparatory Classes for the Grandes Écoles (TSI track) — Sep. 2021 – Jul. 2023

- Two-year intensive programme in mathematics, physics and engineering.

Skills

Programming	Python, C++, MATLAB, R
Machine learning	PyTorch, TensorFlow, scikit-learn
Medical imaging	MONAI, SimpleITK, nibabel
Tools	Docker, Git, Slurm (HPC), Weights & Biases, OpenVINO, Flask
Languages	Arabic: native · French: bilingual · English: C1 (TOEFL iBT 95) · Chinese: A2 (HSK 2)
Volunteering	Tutor for PTSI and PT preparatory classes (100+ hours), Lycée Eugène Livet, Nantes (2023–2024)

Research Projects

Disentangled VAE for ECG Signal Extraction from Degraded Images

Personal research project

- Separated ECG morphology from image artifacts (noise, overlays, blur) in a VAE latent space with LoRA, on 21K+ PTB-XL signals.
- Outperformed median and Gaussian filters by **15–20% in SNR**, preserving P, QRS and T waves; raised arrhythmia F1 by 4–6%.

Physics-Informed Neural Network for Fuel Cell Health Prognosis — *Nantes, France*

LS2N Lab, École Centrale de Nantes, supervised by Mickaël Hilairet

- Encoded the electrochemical degradation dynamics of a hydrogen fuel cell as physics constraints in a neural network loss (PINN).
- Estimated remaining useful life with **24% lower error** than a Kalman filter, on long-term experimental degradation trajectories.

Honors & Awards

2026 **1st Place / 56 Teams**, ENS Paris-Saclay Data Challenge (HrFlow.ai), job recommendation

2025 **3rd Place / 49 Teams**, ENS Paris-Saclay Data Challenge (Raidium), multi-organ CT segmentation